

Question 2:

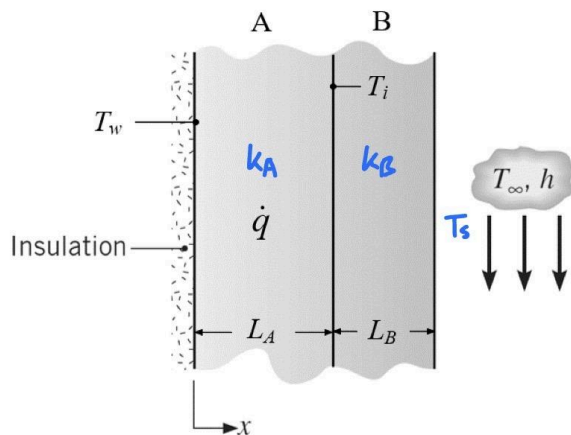
Known:

- Uniform volumetric heat generation rate = $\dot{q}$
- The left surface of Wall A is thermally insulated
- There is no heat generation within Wall B
- The left surface of Wall B is in contact with Wall A
- There is no contact thermal resistance at their boundary
- Coolant temperature: $T_{\infty} = 20^\circ\text{C}$
- Temperature between insulation and Wall A: $T_w = 115^\circ\text{C}$
- Temperature between Wall A and Wall B: $T_i = 100^\circ\text{C}$

Find:

- (A) Calculate the right surface temperature T_s of Wall B
- (B) Calculate the uniform volumetric heat generation $\dot{q}$ at steady state
- (C) Calculate the thermal conductivity k_A of Wall A
- (D) Plot the temperature profile across Wall A and Wall B and show its important features

Schematic:



Properties:

- $L_A = 30\text{ cm}$
- Thermal conductivity of Wall A = k_A
- $L_B = 10\text{ cm}$
- Thermal conductivity of Wall B: $k_B = 30\text{ W/(m}\cdot\text{K)}$
- Convective heat transfer coefficient: $h = 500\text{ W/(m}^2\cdot\text{K)}$

Assumptions:

- Assume 1-D heat transfer
- Assume constant properties

Analysis:

$$\textcircled{a} \quad \dot{E}_{in} + \cancel{\dot{E}_G} - \dot{E}_{out} = \cancel{\dot{E}_S} \rightarrow \dot{E}_{in} = \dot{E}_{out}$$

$$\dot{E}_{in} = k_B \frac{(T_S - T_i)}{L_B} ; \dot{E}_{out} = h(T_S - T_{\infty})$$

$$k_B \frac{(T_S - T_i)}{L_B} = h(T_S - T_{\infty}) \rightarrow (30 \text{ W/mK}) \frac{(T_S - 100^\circ\text{C})}{0.1 \text{ m}} = (500 \text{ W/m}^2\text{K})(T_S - 20^\circ\text{C}) \rightarrow T_S = 50^\circ\text{C}$$

$$\textcircled{b} \quad \cancel{\dot{E}_{in}} + \dot{E}_G - \dot{E}_{out} = \cancel{\dot{E}_S} \rightarrow \dot{E}_G = \dot{E}_{out}$$

$$\dot{E}_G = \dot{q}V ; \dot{E}_{out} = h(T_S - T_{\infty})$$

$$\dot{q}V = h(T_S - T_{\infty}) \rightarrow \dot{q} = \frac{(500 \text{ W/m}^2\text{K})(50^\circ\text{C} - 20^\circ\text{C})}{(0.3 \text{ m})} = 50,000 \text{ W/m}^3$$

$$\textcircled{c} \quad \int \frac{d^2T}{dx^2} = \int \frac{-\dot{q}}{kA} \rightarrow \frac{dT}{dx} = \frac{-\dot{q}}{kA}x + C_1 \quad \left. \frac{dT}{dx} \right|_{x=0} = 0 \rightarrow C_1 = 0$$

$$\int \frac{dT}{dx} = \int \frac{-\dot{q}}{kA}x + C_1 \rightarrow T(x) = \frac{-\dot{q}}{2kA}x^2 + C_1x + C_2$$

$$T(0) = T_v = 115 = C_2$$

$$T(L_A) = T_i = \frac{-\dot{q}}{2kA}L_A^2 + C_2 \rightarrow 100 = -\frac{50000}{2kA}(0.3)^2 + 115 \rightarrow k_A = 150 \text{ W/mK}$$

$$\textcircled{d} \quad T_A(x) = 115 - \frac{\dot{q}}{2k_A}x^2$$

Wall B:

$$\frac{d^2T}{dx^2} = 0$$

$$T_B(x) = C_1x + C_2$$

$$T_B(L_A) = 100 = C_1L_A + C_2$$

$$T_B(L_A + L_B) = 50 = C_1(L_A + L_B) + C_2 \rightarrow C_1 = -500 ; C_2 = 250$$

$$T_B(x) = 250 - 500x$$

